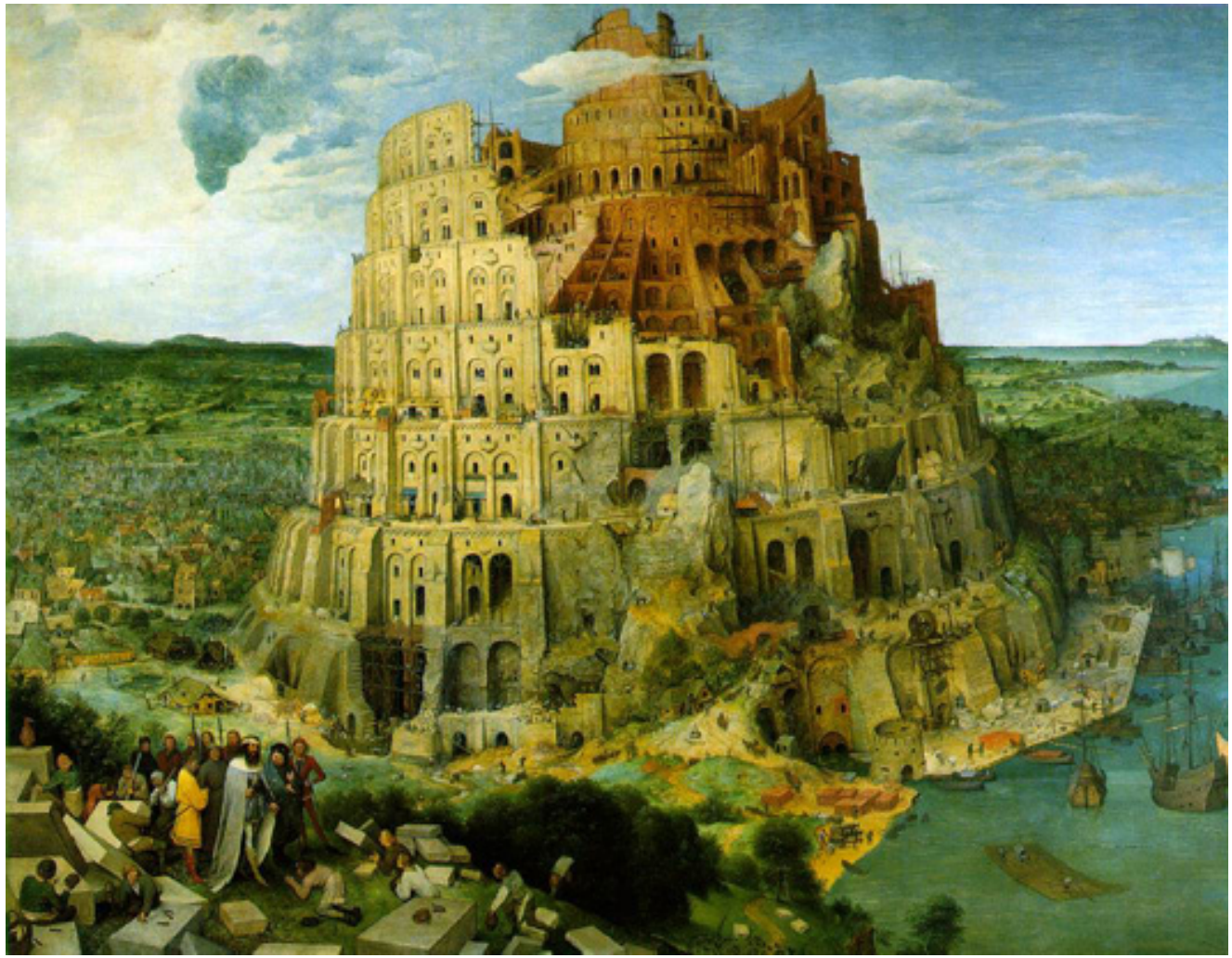
Se busca: The Perfect Language. LIMBER as a ‘metaboundary’ tool.



Supported by:

University of Essex

E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL

Joint Information Systems Committee

UK Data Archive
www.data-archive.ac.uk

IASSIST/IFDO 2001 Amsterdam
Myriam García Bernabé
UK Data Archive
Wednesday, 16th of May, 2001

EU Human Language Technologies IST Project - Language Independent Metadata Browsing of European Resources.

CLRC Rutherford Appleton Laboratory, Intrasoft,
UK Data Archive, Norwegian Data Archive,
user group:- other European national archives

To Breakdown Linguistic and Discipline Barriers to interoperability between European Resources

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

SOMETHING FOR EVERYBODY; WHAT IS LIMBER AS A METABOUNDARY TOOL?

• Multilingual thesaurus for metadata search and access across different languages and cultures.

• Semi-automatic indexing tool for archivists.

• Metadata model (standardisation tool for metadata codebooks, from XML to a more flexible format, RDF?)

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

GOALS OF LIMBER

- Reduce HASSET thesaurus to ELSST thesaurus
- Cull English hierarchies
- Translate reduced hierarchies
- Widen the current coverage
- Put all hierarchies into RDF format
- Use in conjunction with NESSTAR DDI metadata
- Create physical interfaces

* Derive indexing tool from LIMBER thesaurus

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

WHY?

• Broadening horizons we narrow the difficulties. (Broad-based thesaurus)

• ELSST = a common baseline.

• Controlled vocabulary: standardise thesauri content

• Collate user preferences.

• Further cross-national research by resolving ambiguity.

• Allow feedback for the user on keyword search in their own language.

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

PROGRESS TO DATE

• Version 1 ELSST published containing 48 hierarchies: about 1,500 broad terms.

• Some of the hierarchies translated into the target languages (Half into Spanish, quarter into German, a few into French)

• NESSTAR interface and DDI codebook headings translated into all target languages.

• RDF schema ready for ELSST.

• Interfaces in the process of being designed.

• Indexing tool development during the summer (June-September)

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

THE STORY ABOUT THE PERFECT LANGUAGE UNTIL NOW…

• A ‘perfect’ language or sign system that could facilitate cross-boundary language and concepts independently of differences.

. Babel

. ‘Barbaros’

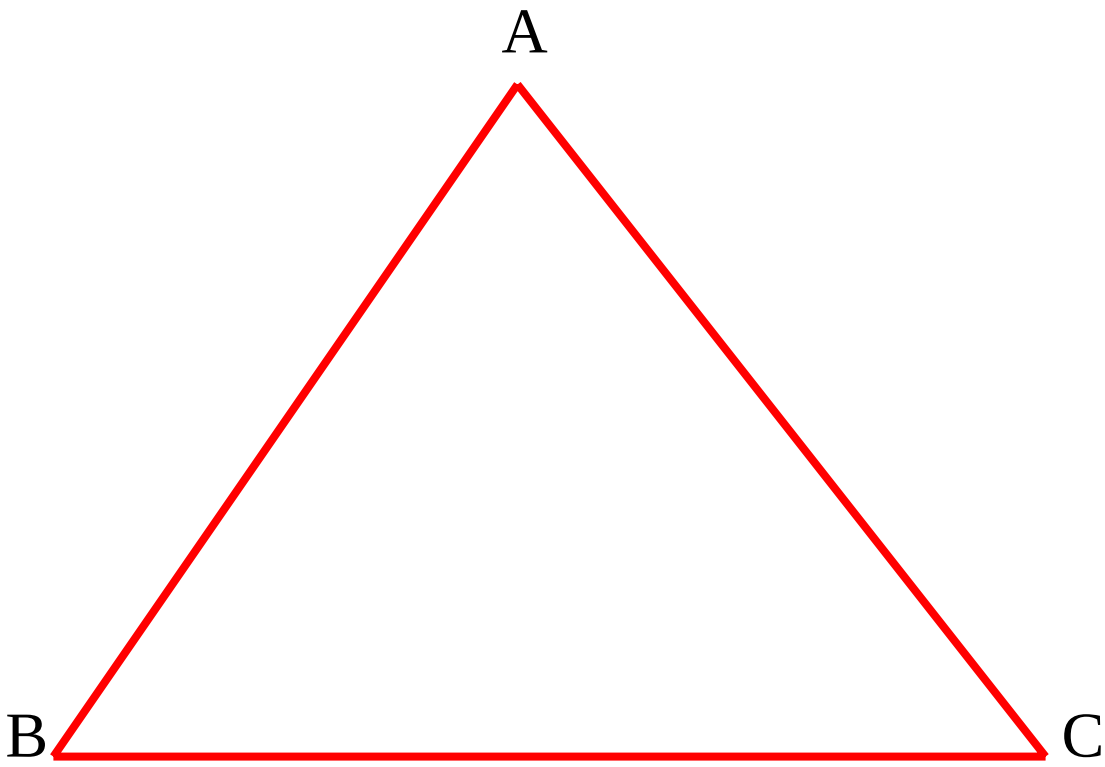
• Most cultures have always obsessed about creating one common language system.

. Esperanto

. Logic, polygraphy, other systems.

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee
UK Data Archive
www.data-archive.ac.uk

THE DREAM



Supported by:
University of Essex
E·S·R·C
ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

WHY A 'METALANGUAGE'?

- Kaleidoscope of languages.

- Linguistic determinism.

- Linguistic relativity.

- Whichever the case, what would make one language better than another one to be used as a basis?

- Metalanguage as a 'super' system

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee
UK Data Archive
www.data-archive.ac.uk

TRANSLATION AND CONCEPTUAL ISSUES

• LIMBER aimed at the presentation of only broad level terms in order to achieve its ‘metaboundary’ objectives.

• Translation of **denotational (referential)** meaning of the terms.

• ISO standard names 5 categories of equivalence. We are only using *exact equivalence*.

•The nominal fallacy.

• *Partial equivalence relationships* may be necessary.

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee
UK Data Archive
www.data-archive.ac.uk

VERBI GRATIA

• Hi-jacking vs. kidnapping = secuestrar (Es.) , entführen (De.) but use of the loanword ‘kidnappen’ **(single-to-multiple equivalence)**

• Joy-riding= 0 **(non-equivalence)**

• Homemaker= sus labores **(partial equivalence)**

• Public housing= vivienda social but not referred to vivienda de protección oficial **(inexact equivalence)**

• Sheltered accommodation = vivienda pública protegida. Student loans = préstamos para estudiantes **(inexact equivalence)**

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

HOW TO RESOLVE THE 'NOT SO EXACT' EQUIVALENCES

- Scope notes for:
  - descriptive noun phrase/ loanword **(single-to-multiple)**
  - non-existent terms/ paraphrasing **(non-equivalence)**
  - partially equivalent terms **(partial equivalence)**
  - ambiguity **(inexact equivalence)**

- Also for the resolution of ambiguity:
  - positioning of terms in hierarchy (controlled vocabulary)
  - Bracketed terms, e.g. Drugs (medicines)/Drugs (illegal)

- Collapsing the ISO equivalencies.

- Technical architecture improvements:
  - elimination of character length limit
  - allowing partial equivalent relationships

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

RDF (Resource Description Framework). THE METALANGUAGE IN LIMBER

• Formal system defined in marked-up text (XML) for the development of **flexible** semantic definitions amongst thesaurus terms.

• RDF provides a machine-understandable model

. Descriptive vocabulary
. Representation of relationships

• Easier to computerise as a metalanguage model, to make it logical. Natural languages are anything but logical!

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

THE METALANGUAGE: AN RDFSCHEMA EXAMPLE

```
<rdf:Description rdf:ID="ES-R73.50/90">
        <rdf:type rdf:resource="http://www.data
archive.ac.uk/Limber/thesaurus#TopConcept"/>
<thes:ClassificationCode>R73.50/90</thes:ClassificationCode>
        <thes:PreferredTerm>
          <rdf:Description>
            <rdf:type rdf:resource="http://www.data
archive.ac.uk/Limber/thesaurus#Term"/>
            <thes:inLanguageOf rdf:resource="http://www.data
archive.ac.uk/Limber/ISO639#es"/>
            <rdf:value>ADICCION</rdf:value>
          </rdf:Description>
        </thes:PreferredTerm>

        <thes:NarrowerConcept rdf:resource="#ES-R73.90"/>
```

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

```
<thes:RelatedConcept rdf:resource="#ES-F54"/>
        <thes:RelatedConcept rdf:resource="#ES-R70/84"/>

        <thes:ExactEquivalent rdf:resource="#EN-R73.50/90">
            <thes:inLanguageOf rdf:resource="http://www.data-
archive.ac.uk/Limber/ISO639#en"/>
        </thes:ExactEquivalent>
        <thes:ExactEquivalent rdf:resource="#FR-R73.50/90">
            <thes:inLanguageOf rdf:resource="http://www.data-
archive.ac.uk/Limber/ISO639#fr"/>
        </thes:ExactEquivalent>
        <thes:ExactEquivalent rdf:resource="#DE-R73.50/90">
            <thes:inLanguageOf rdf:resource="http://www.data-
archive.ac.uk/Limber/ISO639#de"/>
        </thes:ExactEquivalent>
    </rdf:Description>
```

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee

LIMBER AND THE NEW DEVELOPMENTS

• Dream of standard structure for thesauri, mapped through RDF.

• Hopeful signs: translation and acceptance of digital document standards such as Dublin Core, http://dublincore.org/resources/translations/

• Translation of some of the technical specifications for RDF, XML, etc, http://www.w3.org/Consortium/Translation/

• New human and machine interoperability technologies. (DDI, RDF)

• Common goal: universal accessibility and interoperability across boundaries (linguistic, disciplinary, technological).

Supported by:
University of Essex
E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL
Joint Information Systems Committee
UK Data Archive
www.data-archive.ac.uk

REFERENCES

- Matthews, Brian & Miller, Ken. *Having the Right Connections: the Limber Project.*

- Eco, Umberto. *The Search for the Perfect Language*. Blackwell, 1995.

- Pyles, Thomas & Algeo, John. *The Origins and Development of the English Language*. HBJ, 1982, 3rd edition.

- Whorf, B.L. *Language, Thought and Reality*. MIT Press, 1956.

Supported by:

University of Essex

E·S·R·C ECONOMIC & SOCIAL RESEARCH COUNCIL

Joint Information Systems Committee